ters

The cryptographic protocol mirrors the [SIGMA] protocol and uses the Identity Protection Key (IPK) to provide better identity protection. Briefly, the protocol will:

1. Exchange ephemeral elliptic curve public keys (`Sigma1.initiatorEphPubKey` and `Sigma2.responderEphPubKey`) to generate a shared secret
2. Exchange certificates to prove identities (`Sigma2.encrypted2.responderNOC` and `Sigma3.encrypted3.initiatorNOC`)
3. Prove possession of the NOC private key by signing the ephemeral keys and NOC (`sigma-2-tbsdata` and `sigma-3-tbsdata`)

The basic protocol can be achieved within 2 round trips as shown below:

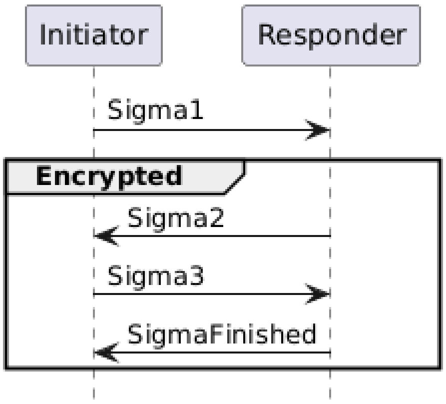

Figure 15. Basic Session Establishment

4.14.2.2. Session Resumption

The protocol also provides a means to quickly resume a session using a previously established session. Resumption does **not** require expensive signature creation and verification which significantly reduces the computation time. Because of this, resumption is favoured for low-powered devices when applicable. Session resumption SHOULD be used by initiators when the necessary state is known to the initiator.

The nomenclature `Sigma1 with Resumption` in the following subsections implies a `Sigma1` message with both the optional `resumptionID` and `initiatorResumeMIC` fields populated in `sigma-1-struct`.

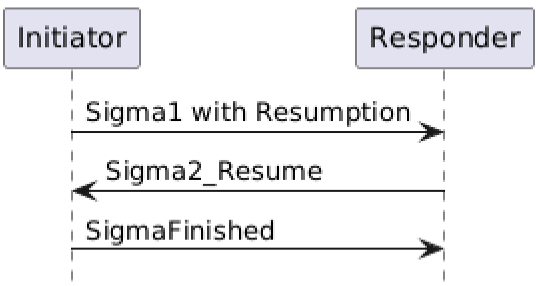

Figure 16. Session Resumption

In the case where a Responder is not able to resume a session as requested by a `Sigma1 with Resumption`, the information included in the `Sigma1 with Resumption` message SHALL be processed as a `Sig-`

